Introduction to GEO5015: Modelling wind and dispersion in urban environments

Clara García-Sánchez, Miguel Martin

Q4 2024-2025

TU Delft

Who am I?

I'm an aerospace engineer

Assistant prof. at 3d geoinformation

Who am I?

I’m an aerospace engineer

Assistant prof. at 3d geoinformation

I’m an amateur…

runner

climber

Who am I?

I’m an aerospace engineer

Assistant prof. at 3d geoinformation

I’m an amateur…

runner

climber

What do I do?

Mainly
Computational
Fluid Dynamics
(CFD)

TU Delft

Who am I?

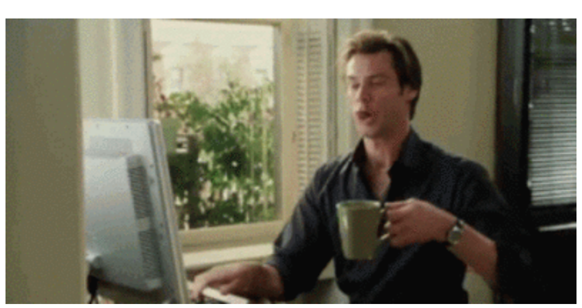
I'm an aerospace engineer

Assistant prof. at 3d geoinformation

I'm an amateur… runner climber

What do I do?

Mainly Computational Fluid Dynamics (CFD)



Who am I?

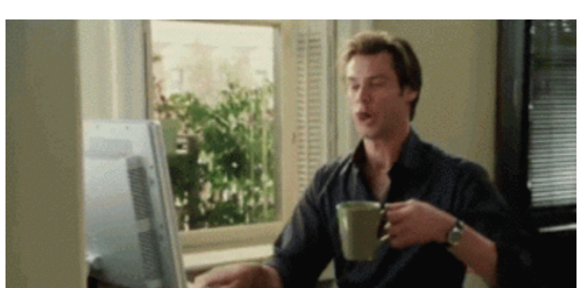
I'm an aerospace engineer

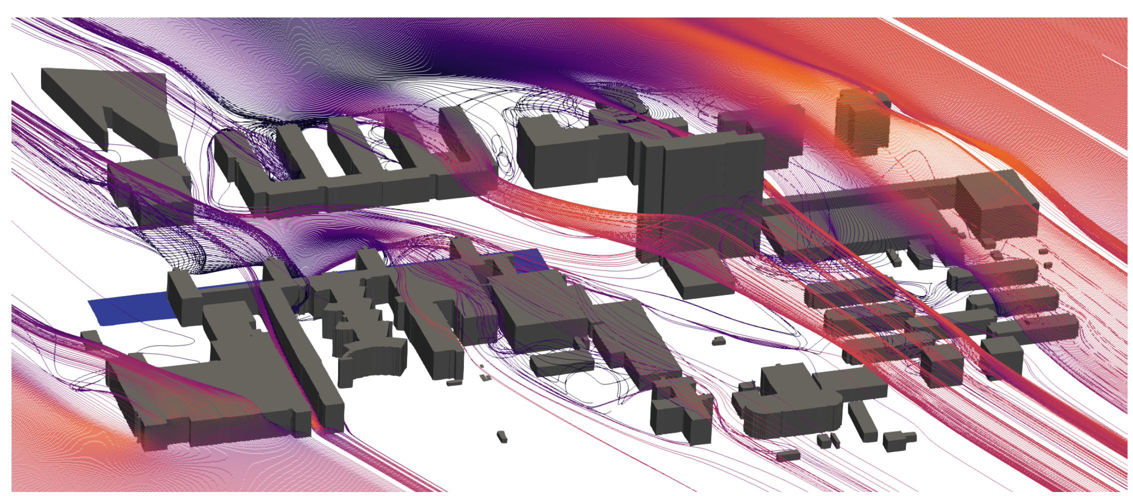
Assistant prof. at 3d geoinformation

I'm an amateur… runner climber

What do I do?

Mainly
Computational
Fluid Dynamics
(CFD)





TU Delft

Who am I?

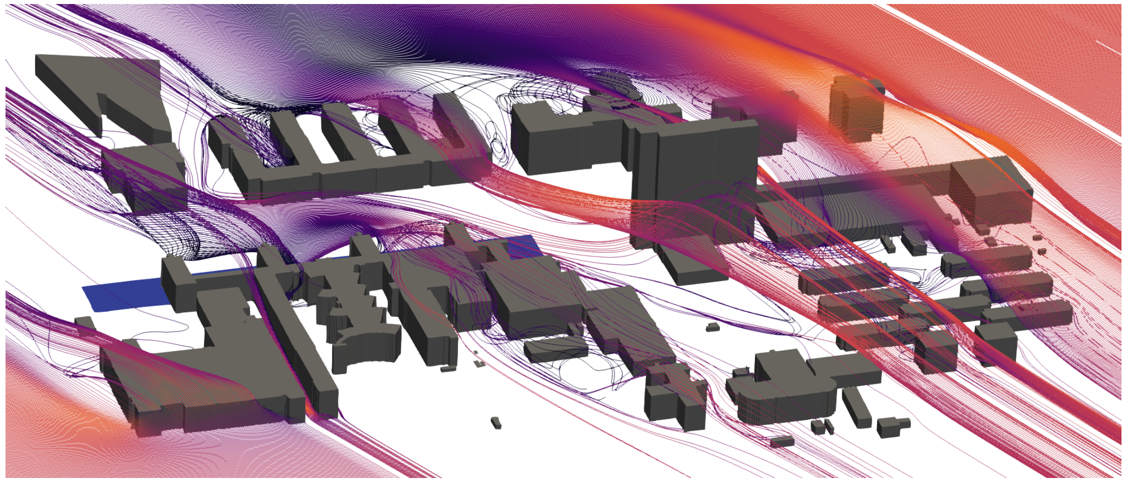
I'm an aerospace engineer

Assistant prof. at 3d geoinformation

I'm an amateur… runner climber

What do I do?

Mainly Computational Fluid Dynamics (CFD)



Wind, temperature and air pollution predictions in urban environments

Nadine Hobeika
PhD candidate

Akshay Patil
Postdoc

Miguel Martin
Postdoc

Learning Objectives:

Understand Computational Fluid Dynamics (CFD)

Understand the physics of the atmospheric boundary layer (ABL)

Apply CFD to a urban case

Determine which physics are important and compromise

In which order do you think we will be completing this LOs?

Or

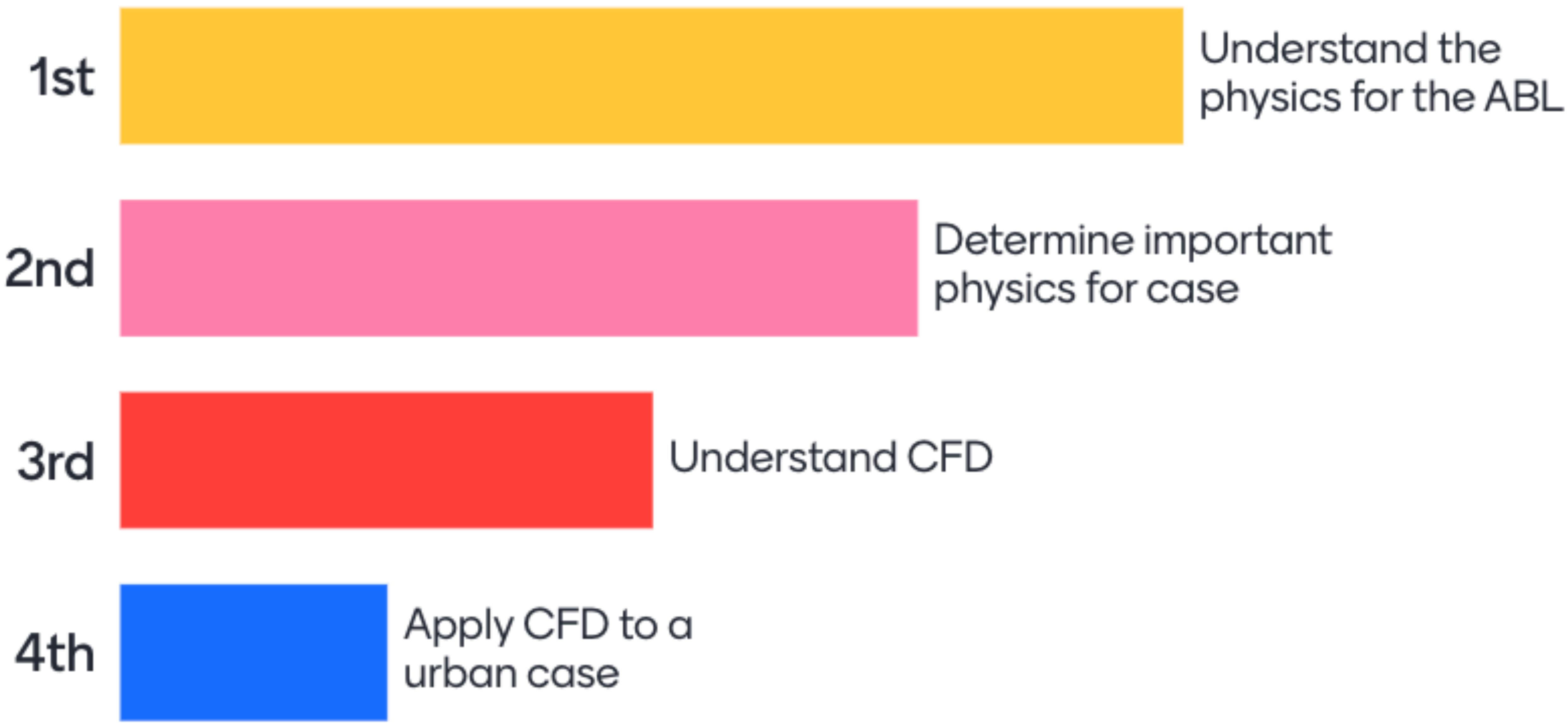
How would you like to complete this LOs?

Applications

Instructions: go to menti.com

Choose the order for the learning objectives that you think it is the correct one!

Learning Objectives:



Learning Objectives:

Understand the physics of the atmospheric boundary layer

Determine which physics are important for each case and know how to compromise

Understand Computational Fluid Dynamics (CFD)

Apply CFD to a urban test case scenario

Classes summary:

Class topic	Week
1. What is Computational Fluid Dynamics?	1
2. An introduction to the atmospheric boundary layer	2
3. Some mathematical and conceptual tools	2
4. Computational Fluid Dynamics applied to urban cases	3
5. Governing equations for turbulent flow	4
6. Turbulent kinetic energy & stability	5
7. The turbulence closure problem	6
8. Support	7,8,9

Calendar:

Date	Hour	Hall	Type class
22/04	8:45-10:45	T	Theory 1
24/04	CANCELLED - STRIKE SUPPORT		
28/04	10:45-12:45	T	Theory 2
29/04	8:45-10:45	T	Theory 2
1/05	8:45-10:45	Geolab	Laboratory
5/05	HOLIDAY		
6/05	8:45-10:45	T	Theory 3
8/05	8:45-10:45	Geolab	Laboratory
12/05	10:45-12:45	T	Theory 4
13/05	8:45-10:45	T	Theory 4
15/05	8:45-10:45	Geolab	Laboratory
19/05	10:45-12:45	T	Theory 5
20/05	8:45-10:45	C	Laboratory
22/05	8:45-10:45	Geolab	Theory 5
26/05	10:45-12:45	F	Theory 6
27/05	8:45-10:45	F	Laboratory
29/05	HOLIDAY		
2/06	10:45-12:45	T	Theory 7
3/06	8:45-10:45	T	Laboratory
5/06	8:45-10:45	Geolab	Laboratory
9/06	HOLIDAY		
10/06	8:45-10:45	T	Laboratory
12/06	8:45-10:45	Geolab	Laboratory

What do I expect from you:

1. Ask questions

2. Be critical with your CFD decisions

3. Collaborate with your peers

4. Submit all your assignments

Too fast…
Something unclear…
You can't hear me…
In my classes there aren't stupid questions

What do I expect from you:

1. Ask questions

2. Be critical with your CFD decisions

3. Collaborate with your peers

4. Submit all your assignments

Too fast…
Something unclear…
You can't hear me…
In my classes there aren't stupid questions

This class if for **you**, I will learn something for sure, but it is you who should get most of the learning!!

What do I expect from you:

1. Ask questions

2. Be critical with your CFD decisions

3. Collaborate with your peers

4. Submit all your assignments

Too fast…
Something unclear…
You can't hear me…
In my classes there aren't stupid questions

This class if for **you**, I will learn something for sure, but it is you who should get most of the learning!!

<u>Classes are for:</u>
Asking, discussing, thinking, doubting, making mistakes, learning

What do you expect from me?:

1. Answer questions (that I know)

2. Give you feedback on time (no later than a week after submission)

3. ....?

What should you NOT expect from me?:

1. That I run your cases

2. That I give you the answers to the questions in the assignment

3. That I read the materials that were proposed for you to read

About the assessment !!!!

What: TWO assignments:

1. Individual (25%) —> can be completed after 3rd week teaching

2. Groups of 2 (75%) —> most of it can be completed after 5th week teaching

Both assignments need to be submitted to pass the class!!
Minimum grade to pass an assignment is 5.0, minimum grade to pass the course is 6.0

When: 23th of June latest at 17:00

Resit: 11th of July at 17:00 (different test case!!)

Where: posted in Brightspace, submitted through Brightspace

Grades: one week later than latest submission date, so 30th of June (if you submit your 1st assignment I can give you the grade earlier, but not the details about it)

Introduce yourself

Instructions to your peer: introduce yourself, why did you enrol this class? What do you expect to learn?